

Applications

Flame photometry is primarily used for the analysis of those elements which have an easily excited flame spectrum of sufficient intensity for detection by a photocell.

For example, for the analysis of sodium, potassium, calcium and lithium.

13 PHOTO ELECTRON SPECTROSCOPY

Photo electron spectroscopy is the experimental technique for the determination of atomic and molecular energy levels.

It basically consists of study of photoelectrons ejected from atoms or molecules when these are bombarded with photons from monochromatic source.

When a photon of frequency ν is incident on atom or molecule, its energy ($h\nu$) is transferred to the electrons.

(A) *In the case of atoms*, part of the energy is consumed to overcome the coulomb forces between electron and nucleus, called binding energy (equal to ionization energy in the case of metals during the study of photoelectric effect). Remaining energy of the photon is released in the form of kinetic energy ($= \frac{1}{2}mv^2$) of emitted electron.

$$\text{Thus, } h\nu = \frac{1}{2}mv^2 + \text{B.E.} \quad \dots(1)$$

$$\Rightarrow \frac{1}{2}mv^2 = h\nu - \text{B.E.} \quad \dots(2)$$

(B) *In the case of molecules*, part of photon's energy is also utilized for the excitation of electrons to higher vibrational or rotational energy levels.

$$\text{Thus, } h\nu = \frac{1}{2}mv^2 + E_{\text{vib./rot.}} + \text{B.E.} \quad \dots(3)$$

$$\frac{1}{2}mv^2 = h\nu - E_{\text{vib./rot.}} - \text{BE} \quad \dots(4)$$

The binding energy of valence electrons is less than that of core electrons. Moreover, B.E.'s are different for different atoms and molecules and so is the $E_{\text{vib./rot.}}$ (for molecules). Thus, according to equations (2) and (4), the kinetic energies of emitted photoelectrons will be different

- (a) for different atoms or molecules.
- (b) for valence electrons and core electrons.

13.1 The Principle of Photoelectron Spectroscopy

An incoming photon (of known energy) collides with an electron in one of the orbitals and expels it with a kinetic energy that is equal to the difference between the energy supplied by the photon and the ionization energy from the occupied orbital.

An electron from an orbital with a low ionization energy will emerge with a high kinetic energy, while an electron from an orbital with a high ionization energy will emerge with a low kinetic energy.

Schrodinger Egn

Consider the simplest type of wave motion
vibration of a stretched string. Then

$$\frac{\partial^2 \psi}{\partial x^2} = -F(x) \quad [-A, A]$$

An e^- from an orbital with low IE will emerge with high KE whereas an e^- from an orbital with a high IE will be ejected with low KE.

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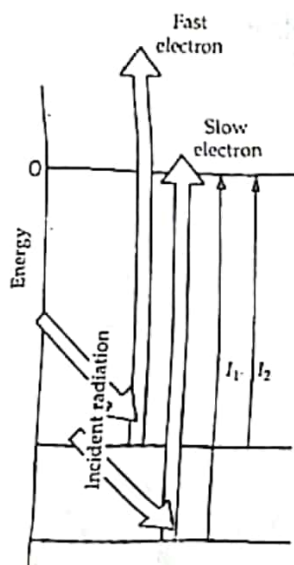


Fig. 57. The basic principle of photoelectron spectroscopy.

These ejected photoelectrons with different kinetic energies are passed through charged plates of electrostatic deflectors. As their kinetic energies are different so is the extent of deflection. Hence, the ejected photoelectrons are separated according to their kinetic energies. The photoelectron spectrum is a record of the number of electrons detected at a particular energy, per unit time.

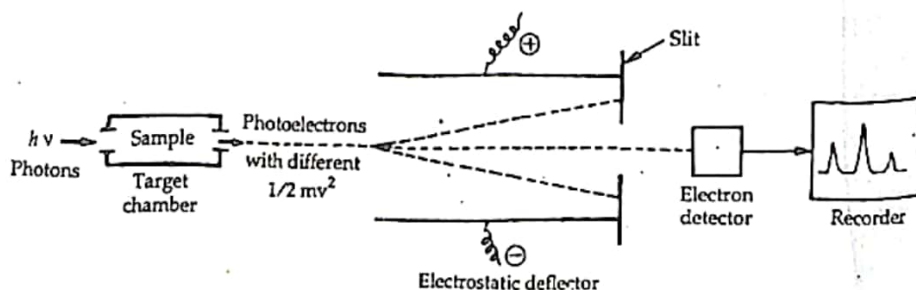


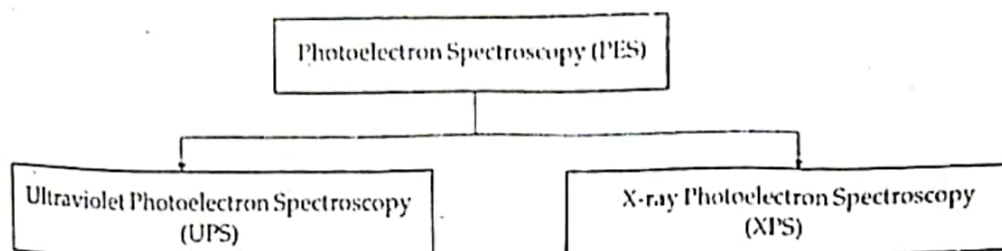
Fig. 58. Photoelectron Spectrometer.

In the case of atoms, Photoelectron spectrum shows as many peaks as are the number of shells viz. valence shells and core-shells. Each peak corresponds to the ionization from a single atomic orbital.

In the case of molecules, Photoelectron spectrum contains many vibrational and rotational lines for each type of electron ionized and thus a band results for the ionization of electron from a single molecular orbital. The number of bands are equal to the number of occupied molecular orbitals.

13.2 Types of Photoelectron Spectroscopy

Broadly speaking, photoelectron spectroscopy is of following two types :



These techniques are briefly discussed below :

(i) **Ultra-violet Photoelectron Spectroscopy (UPS)** : When valence electrons are being studied, then it is essential to work in at least the ultraviolet region of the spectrum with wavelengths of less than 200 nm (because the ionization energies of molecules are several electron volts even for valence electrons). That's why, mostly work is being done with radiation generated by a discharge through Helium.

The He(I) line ($1s^1 2p^1 \rightarrow 1s^2$) lies at 58.43 nm, corresponding to a photon energy of 21.22 eV. Its use gives rise to the technique of ultraviolet photoelectron spectroscopy.

Example 1. Interpret the following ultraviolet photoelectron spectrum of HBr.

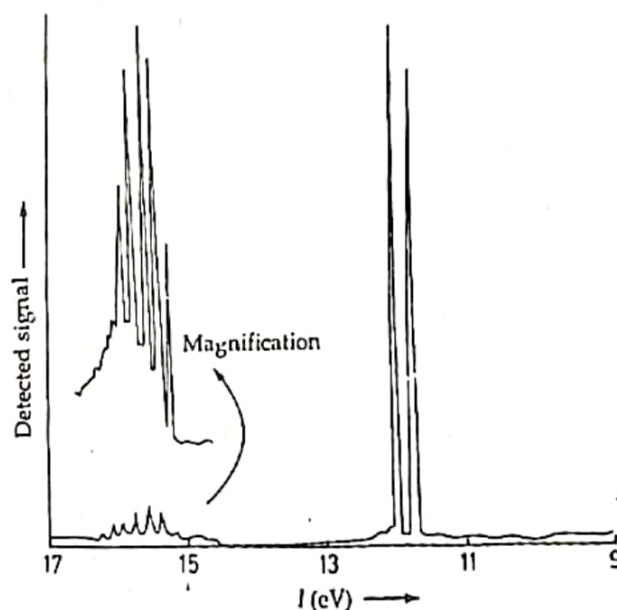


Fig. 59. The photoelectron spectrum of HBr.

Solution. The spectrum has two main groups (disregard the fine structure). The lowest ionization energy band corresponds to the ionization of a Br lone-pair electron. Because these electrons in the lone pairs of Br atom are least tightly bound so they have highest kinetic energies when ejected. The lack of much vibrational structure in this band is due to the fact that the lone pair electrons are at the equilibrium distance from the nucleus.

of an electron from the HBr σ bond. This bond shows the presence of much vibrational structure. It confirms that ejection of a σ electron is accompanied by a considerable amount of vibrational excitation of HBr^+ that results from the ionization. As σ electron is lost, we can expect a considerable decrease of equilibrium bond length between HBr and HBr^+ .

Example 2.(a) The highest kinetic energy electrons in the spectrum of H_2O using 21.22 eV radiation are at about 9 eV and show a large vibrational spacing of 0.41 eV. The symmetric stretching mode of the neutral H_2O molecule lies at 3652 cm^{-1} . What conclusion can be drawn from the nature of the orbital from which the electron is ejected?

(b) In the same photoelectron spectrum of H_2O , the band near 7.0 eV shows a long vibrational series with spacing 0.125 eV. The bending mode of H_2O lies at 1596 cm^{-1} . What conclusions can you draw about the characteristics of the orbital occupied by the photoelectron?

Solution. The ejected electron had little influence on bonding in the molecule, if the vibrational separation in the ion is similar to that in the molecule. However, presence of a lot of vibrational structure indicates that the electron had been heavily involved in bonding.

(a) Since $1\text{ eV} = 8065.5\text{ cm}^{-1}$

$\therefore 0.41\text{ eV} = 3306.9\text{ cm}^{-1}$.

This is similar to the 3652 cm^{-1} of neutral H_2O molecule. It means the electron is ejected from an orbital that has little influence on the bonding in the molecule. Thus, the nature of the orbital from which the electron was ejected is 'non-bonding'.

(b) Since $1\text{ eV} = 8065.5\text{ cm}^{-1}$

$\therefore 0.125\text{ eV} = 1008.2\text{ cm}^{-1}$.

This is not similar to the 1596 cm^{-1} corresponding to the bending mode of H_2O . The electron contributes to the long-distance HH bonding across the molecule.

(ii) **X-ray Photoelectron Spectroscopy (XPS)** : When the aim is to study core electrons, photons of higher energy are needed to expel them : X-rays are used, and the technique is known as X-ray photoelectron spectroscopy (XPS).

The XPS gives lines characteristic of the elements present in a compound or alloy. This is because, with XPS, we are studying core electrons and their ionization energies are insensitive to the bonds between atoms (as they are very tightly bound to be greatly affected by the changes that accompany bond formation) so core ionization energies are characteristic of the individual atom rather than the overall molecule. The K-shell ionization energies of few elements are given below :

Element	Li	Be	B	C	N	O	F
The K-shell Ionization energy (eV)	50	110	190	280	400	530	690

Detection of one of these values (and values corresponding to ejection from other inner shells) indicates the presence of the corresponding element. This application is responsible for the alternative name electron spectroscopy for chemical analysis (ESCA).

Limitation. Even though X-ray may penetrate into the bulk sample, the ejected electrons cannot escape except from within a few nanometers of the surface. Thus, XPS is mainly limited to the study of surface layers.

Applications. XPS is a very useful technique for studying the surface state of heterogeneous catalysts, the differences between surface and bulk structures, and the processes that can cause damage to high-temperature superconductors and semiconductor wafers. XPS can also be used to obtain valuable information about the presence of chemically inequivalent atoms of the same molecule. For instance, the XPS spectrum of azide ion, N_3^- , gives the spectrum which lies at 400 eV (and hence is typical of N 1s electrons). It has a doublet spectrum with splitting. For explaining the doublet structure with spacing, we note that structure of the azide

ions is $\overset{\ominus}{\text{N}} = \overset{\oplus}{\text{N}} = \overset{\ominus}{\text{N}}$. The presence of the negative charges on the terminal nitrogen atoms lowers the core ionization energies, whereas the positive charge on the central atom raises it. This inequivalence of N atoms in N_3^- ion results in two lines in the spectrum with intensities in the ratio 2:1.

13.3 Applications of Photoelectron Spectroscopy

1. Determination of atomic and molecular energy levels.
2. Determination of ionization energies of valence and core electrons of atoms and molecules.

13.4 Limitations

Sometimes interpretation of Photoelectron spectrum is not straight-forward. Because number of bands appearing in the spectrum are more than the number of orbitals present in a molecule. For instance, the ionization from degenerate occupied molecular orbital can give rise to many bands in the spectrum. This is because once we remove one electron from degenerate molecular orbital, the degeneracy is lost in the resulting positive ion.

14 MASS SPECTROMETRY

14.1 Introduction

In the *mass spectrometer* (figure 60), molecules are bombarded with a beam of energetic electrons and converted to highly energetic positively charged ions (molecular ions or *parent ions*), which can break up into smaller ions (*daughter ions*, or fragment ions). The loss of an electron from a molecule leads to the formation of a radical cation ($M \longrightarrow M^+$). The molecular ion M^+ further decomposes to a pair of fragments, which may be either a radical plus an ion ($M^+ \longrightarrow m_1^+ + m_2^\cdot$), or a small molecule plus a radical cation ($M^+ \longrightarrow m_1 + m_2^+$). The molecular ions, the fragment ions and the fragment radical ions are separated by deflection in a variable magnetic field in proportion of mass to charge, m/z . They also generate a